



# Low-complexity 8-point DCT approximation based on angle similarity for image and video coding

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## Abstract

The principal component analysis (PCA) is widely used for data decorrelation and dimensionality reduction. However, the use of PCA may be impractical in real-time applications, or in situations where energy and computing constraints are severe. In this context, the discrete cosine transform (DCT) becomes a low-cost alternative to data decorrelation. This paper presents a method to derive computationally efficient approximations to the DCT. The proposed method aims at the minimization of the angle between the rows of the exact DCT matrix and the rows of the approximated transformation matrix. The resulting transformation matrices are orthogonal and have extremely low arithmetic complexity. Considering popular performance measures, one of the proposed transformation matrices outperforms the best competitors in both matrix error and coding capabilities. Practical applications in image and video coding demonstrate the relevance of the proposed transformation. *In fact, we show that the proposed approximate DCT can outperform the exact DCT for image encoding under certain compression ratios.* The proposed transform and its direct competitors are also physically realized as digital prototype circuits using FPGA technology.

**Keywords** DCT approximation · HEVC · Compression

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# 1 Introduction

Data decorrelation is a central task in many statistical and signal processing problems (Dunteman 1989; Jolliffe 2002; Britanak et al. 2007). Decorrelation can be accomplished by means of a linear transformation that converts correlated observations into linearly uncorrelated values. This operation is commonly performed by principal component analysis (PCA) (Jolliffe 2002). PCA is widely used to reduce the dimensionality of data (Jolliffe 2002; Gorban et al. 2007), where the information contained in all the original variables is replaced by the data variability information of the initial few uncorrelated principal components. The quality of such approximation depends on the number of components used and the proportion of variance explained, or energy retained, by each of them.

In the field of analysis and processing of images and signals, PCA, also known as the discrete Karhunen–Loève transform (KLT) (Britanak et al. 2007), is considered the optimal linear transformation for data decorrelation when the signal is a first order Markov process (Rao and Yip 1990; Britanak et al. 2007). The KLT has the following features (Britanak et al. 2007): (i) decorrelates the input data completely in the transform domain; (ii) minimizes the mean square error in data compression; and (iii) concentrates the energy (variance) in a few coefficients of the output vector. Because the KLT matrix depends on the variance and covariance matrix of the input data, deriving computationally efficient algorithms for real-time processing becomes a very hard task (Chen and Zeng 2012; Ivarez-Cortés et al. 2017; Bae and Yoo 2017; Thomakos 2016; Zeng et al. 2017; Blahut 2010; Ahmed et al. 1974; Britanak et al. 2007; Clarke 1981).

If the input data follows a stationary highly correlated first-order Markov process (Ahmed et al. 1974; Britanak et al. 2007; Cintra et al. 2014), then the KLT is very closely approximated by the discrete cosine transform (DCT) (Ahmed et al. 1974; Britanak et al. 2007). Natural images fall into this particular statistical model category (Gonzalez and Woods 2006). Thus DCT inherits the decorrelation and compaction properties of the KLT, with the advantage of having a closed-form expression independent of the input signal. Image and video communities widely adopt the DCT in their most successful compression standards, such as JPEG (Wallace 1992) and MPEG (Puri 2004). Often such standards include two-dimensional (2-D) versions of the DCT applied to small image blocks ranging from  $4 \times 4$  to  $32 \times 32$  pixels.

The  $8 \times 8$  block is employed in a large number of standards, for example: JPEG (Wallace 1992), MPEG (Gall 1992), H.261 (International Telecommunication Union 1990), H.263 (International Telecommunication Union 1995), H.264/AVC (Luthra et al. 2003), and HEVC (Pourazad et al. 2012). The arithmetic cost of the 8-point DCT is 64 multiplications and 56 additions, when computed by definition. Fast algorithms can dramatically reduce the arithmetic cost to 11 multiplications and 29 additions, as in the Loeffler DCT algorithm (Loeffler et al. 1989).

However, the number of DCT calls in common image and video encoders is extraordinarily high. For instance, a single image frame of high-definition TV (HDTV) contains 32,400  $8 \times 8$  image subblocks. Therefore, computational savings in the DCT step may effect significant performance gains, both in terms of speed and power consumption (Potluri et al. 2014; Coutinho et al. 2015). Being quite a mature area of research (Bayer et al. 2012), there is little room for improvement on the exact DCT computation. Thus, one approach to further minimize the computational cost of computing the DCT is the use of matrix approximations (Bayer and Cintra 2010; Cintra et al. 2014). Such approximations provide matrices with similar mathematical behavior to the exact DCT while presenting a dramatically low arithmetic cost.

The goals of this paper are as follows. First, we aim at establishing an optimization problem to facilitate the derivation of 8-point DCT approximations. To this end, we adopt a vector angle based objective function to minimize the angle between the rows of the approximate and the exact DCT matrices subject to orthogonality constraints. Second, the sought approximations are (i) submitted to a comprehensive assessment based on well-known figures of merit and (ii) compared to state-of-the-art DCT approximations. Third, fast algorithms are derived and realized in FPGA hardware with comparisons with competing methods. We also examine the performance of the obtained transformations in the context of image compression and video coding. *We demonstrate that one of our DCT approximations can outperform the exact DCT in terms of effected quality after image compression.*

This paper is organized as follows. In Sect. 2, the 8-point DCT and popular DCT approximations are discussed. Section 3 introduces an optimization problem to pave the way for the derivation of new DCT approximations. In Sect. 4 the proposed approximations are detailed and assessed according to well-known performance measures. In Sect. 5 a fast algorithm for the proposed approximation is presented. Moreover, a field-programmable gate array (FPGA) design is proposed and compared with competing methods. Section 6 furnishes computational evidence of the appropriateness of the introduced approximate DCT for image and video encoding. Section 7 concludes the paper.

## 2 DCT approximations

Let  $\mathbf{x}$  and  $\mathbf{X}$  be 8-point column vectors related by the DCT. Therefore, they satisfy the following expression:  $\mathbf{X} = \mathbf{C} \cdot \mathbf{x}$ , where

$$\mathbf{C} = \begin{bmatrix} \gamma_3 & \gamma_3 & \gamma_3 & \gamma_3 & \gamma_3 & \gamma_3 & \gamma_3 & \gamma_3 \\ \gamma_0 & \gamma_2 & \gamma_4 & \gamma_6 & -\gamma_6 & -\gamma_4 & -\gamma_2 & -\gamma_0 \\ \gamma_1 & \gamma_5 & -\gamma_5 & -\gamma_1 & -\gamma_1 & -\gamma_5 & \gamma_5 & \gamma_1 \\ \gamma_2 & -\gamma_6 & -\gamma_0 & -\gamma_4 & \gamma_4 & \gamma_0 & \gamma_6 & -\gamma_2 \\ \gamma_3 & -\gamma_3 & -\gamma_3 & \gamma_3 & \gamma_3 & -\gamma_3 & -\gamma_3 & \gamma_3 \\ \gamma_4 & -\gamma_0 & \gamma_6 & \gamma_2 & -\gamma_2 & -\gamma_6 & \gamma_0 & -\gamma_4 \\ \gamma_5 & -\gamma_1 & \gamma_1 & -\gamma_5 & -\gamma_5 & \gamma_1 & -\gamma_1 & \gamma_5 \\ \gamma_6 & -\gamma_4 & \gamma_2 & -\gamma_0 & \gamma_0 & -\gamma_2 & \gamma_4 & -\gamma_6 \end{bmatrix},$$

and  $\gamma_k = \cos(2\pi(k+1)/32)$ , for  $k = 0, 1, \dots, 6$ .

Common algorithms for efficient DCT computation include: (i) Yuan et al. (2006), (ii) Arai et al. (1988), (iii) Chen et al. (1977), (iv) Feig and Winograd (1992), (v) Lee (1984), and (vi) Hou (1987). Table 1 lists the computational costs associated to such methods. The theoretical minimal multiplicative complexity is 11 multiplications (Heideman and Burrus 1988; Loeffler et al. 1989), which is attained by the Loeffler algorithm (Loeffler et al. 1989).

A DCT approximation is a matrix  $\hat{\mathbf{C}}$  capable of furnishing  $\hat{\mathbf{X}} = \hat{\mathbf{C}} \cdot \mathbf{x}$  where  $\hat{\mathbf{X}} \approx \mathbf{X}$  according to some prescribed criterion, such as matrix proximity or coding performance (Britanak et al. 2007). In general terms, as shown in Higham (1986), Cintra and Bayer (2011), Cintra (2011), Britanak et al. (2007), Bayer and Cintra (2012),  $\hat{\mathbf{C}}$  is a real valued matrix which consists of two components: (i) a low-complexity matrix  $\mathbf{T}$  and (ii) a diagonal matrix  $\mathbf{S}$ . Such matrices are given by:

$$\hat{\mathbf{C}} = \mathbf{S} \cdot \mathbf{T}, \quad (1)$$

where

$$\mathbf{S} = \sqrt{(\mathbf{T} \cdot \mathbf{T}^T)^{-1}}. \quad (2)$$

**Table 1** Computational cost of the fast algorithms for the DCT

| Algorithm                                                               | Multiplications | Additions |
|-------------------------------------------------------------------------|-----------------|-----------|
| Loeffler et al. (1989), Liang and Tran (2001) and Masera et al. (2017a) | 11              | 29        |
| Yuan et al. (2006) and Wang (2011)                                      | 12              | 29        |
| Lee (1984), Snigdha et al. (2016), Suzuki and Ikehara (2010)            | 12              | 29        |
| Hou (1987), Fong and Cham (2012) and Choi et al. (2010)                 | 12              | 29        |
| Arai et al. (1988), Liang and Tran (2001) and Masera et al. (2017b)     | 13              | 29        |
| Chen et al. (1977), Park et al. (2012) and Liang and Tran (2001)        | 16              | 26        |
| Feig–Winograd (1992, 2016)                                              | 22              | 28        |

The operation  $\sqrt{\cdot}$  is the matrix square root operation (Higham 1987; Seber 2008).

Hereafter *low-complexity* matrices are referred to as  $\mathbf{T}_*$ , where the subscript  $*$  indicates the considered method. Also *DCT approximations* are referred to as  $\hat{\mathbf{C}}_*$ . If the subscript is absent, then we refer to a generic low-complexity matrix or DCT approximation.

A traditional DCT approximation is the signed DCT (SDCT) (Haweel 2001) which matrix is obtained according to:  $\frac{1}{\sqrt{8}} \cdot \text{sgn}(\mathbf{C})$ , where  $\text{sgn}(\cdot)$  is the entry-wise signum function. Therefore, in this case, the entries of the associated low-complexity matrix  $\mathbf{T}_{\text{SDCT}} = \text{sgn}(\mathbf{C})$  are in  $\{0, \pm 1\}$ . Thus matrix  $\mathbf{T}_{\text{SDCT}}$  is multiplierless.

Notably, in the past few years, several approximations for the DCT have been proposed as, for example, the rounded DCT (RDCT,  $\mathbf{T}_{\text{RDCT}}$ ) (Bayer and Cintra 2010), the modified RDCT (MRDCT,  $\mathbf{T}_{\text{MRDCT}}$ ) (Bayer and Cintra 2012), the series of approximations proposed by Bouguezel–Ahmad–Swamy (BAS) (Bouguezel et al. 2008, 2009, 2011, 2013), the Lengwehasatit–Ortega approximation (LO,  $\mathbf{T}_{\text{LO}}$ ) (Lengwehasatit and Ortega 2004), the approximation proposed by Senapati et al. (2010), and the collection of approximations introduced in Cintra et al. (2014). Most of these approximations are orthogonal with low computational complexity matrix entries. Essentially, they are matrices defined over the set  $\{0, \pm 1/2, \pm 1, \pm 2\}$ , with the multiplication by powers of two implying simple bit-shifting operations.

Such approximations were demonstrated to be competitive substitutes for the DCT and its related integer transformations as shown in Bayer and Cintra (2010, 2012), Bouguezel et al. (2008, 2009, 2011, 2013), Lengwehasatit and Ortega (2004) and Cintra et al. (2014). Table 2 illustrates some common integer transformations linked to the DCT approximations.

### 3 Greedy approximations

#### 3.1 Optimization approach

Approximate DCT matrices are often obtained by fully considering the exact DCT matrix  $\mathbf{C}$ , including its symmetries (Cham 1989), fast algorithms (Hou 1987; Loeffler et al. 1989), parametrizations (Feig and Winograd 1992), and numerical properties (Yuan et al. 2006). Usually the low-complexity component of a DCT approximation is found by solving the following optimization problem:

**Table 2** Common 8-point low-complexity matrices associated to DCT approximations

| Method                                                   | Transformation Matrix                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{T}_{\text{RDCT}}$ (Bayer and Cintra 2010)       | $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 \\ 1 & 0 & 0 & -1 & -1 & 0 & 0 & 1 \\ 1 & 0 & -1 & -1 & 1 & 1 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & -1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & -1 & 1 & -1 & 1 & 0 \end{bmatrix}$                                                                                     |
| $\mathbf{T}_{\text{BAS-2008b}}$ (Bouguezel et al. 2008)  | $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 0 & -1 & 0 & 0 & 1 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & 0 & 0 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix}$                                                                                 |
| $\mathbf{T}_{\text{LO}}$ (Lengwehasatit and Ortega 2004) | $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 \\ 1 & \frac{1}{2} & -\frac{1}{2} & -1 & -1 & -\frac{1}{2} & \frac{1}{2} & 1 \\ 1 & 0 & -1 & -1 & 1 & 1 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 \\ \frac{1}{2} & -1 & 1 & -\frac{1}{2} & -\frac{1}{2} & 1 & -1 & \frac{1}{2} \\ 0 & -1 & 1 & -1 & 1 & -1 & 1 & 0 \end{bmatrix}$ |
| $\mathbf{T}_6$ (Cintra et al. 2014)                      | $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ 2 & 1 & 1 & 0 & 0 & -1 & -1 & -2 \\ 2 & 1 & -1 & -2 & -2 & -1 & 1 & 2 \\ 1 & 0 & -2 & -1 & 1 & 2 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -2 & 0 & 1 & -1 & 0 & 2 & -1 \\ 1 & -2 & 2 & -1 & -1 & 2 & -2 & 1 \\ 0 & -1 & 1 & -2 & 2 & -1 & 1 & 0 \end{bmatrix}$                                                                                 |
| $\mathbf{T}_4$ (Cintra et al. 2014)                      | $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 0 & -1 & -1 & 1 & 1 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 0 & -1 & 1 & -1 & 1 & -1 & 1 & 0 \end{bmatrix}$                                                                                 |

$$\mathbf{T} = \arg \min_{\mathbf{T}'} \text{approx}(\mathbf{T}', \mathbf{C}),$$

where  $\text{approx}(\cdot, \cdot)$  is a particular approximation assessment function—such as proximity measures and coding performance metrics (Britanak et al. 2007)—and subject to various constraints, such as orthogonality and low-complexity of the candidate matrices  $\mathbf{T}'$ .

However, the DCT matrix can be understood as a stack of row vectors  $\mathbf{c}_k^T, k = 1, 2, \dots, 8$ , as follows:

$$\mathbf{C} = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{c}_3 \quad \mathbf{c}_4 \quad \mathbf{c}_5 \quad \mathbf{c}_6 \quad \mathbf{c}_7 \quad \mathbf{c}_8]^T. \quad (3)$$

In the current work, to derive an approximation for  $\mathbf{C}$ , we propose to individually approximate each of its rows in the hope that the set of approximate rows generate a good approximate matrix. Such heuristic can be categorized as a greedy method (Cormen et al. 2001). Therefore, our goal is to derive a low-complexity integer matrix

$$\mathbf{T} = [\mathbf{t}_1 \quad \mathbf{t}_2 \quad \mathbf{t}_3 \quad \mathbf{t}_4 \quad \mathbf{t}_5 \quad \mathbf{t}_6 \quad \mathbf{t}_7 \quad \mathbf{t}_8]^T \quad (4)$$

such that its rows  $\mathbf{t}_k^T, k = 1, 2, \dots, 8$ , satisfy:

$$\mathbf{t}_k = \arg \min_{\mathbf{t} \in \mathcal{D}} \text{error}(\mathbf{t}, \mathbf{c}_k), \quad k = 1, 2, \dots, 8, \quad (5)$$

**Table 3** Examples of approximated vectors for the search space  $\mathcal{D}_1$ 

| $n$      | Approximated vector                            |
|----------|------------------------------------------------|
| 1        | $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]^\top$         |
| 2        | $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ -1]^\top$        |
| 3        | $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 0]^\top$         |
| 4        | $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ -1 \ 1]^\top$        |
| $\vdots$ | $\vdots$                                       |
| 6558     | $[-1 \ -1 \ -1 \ -1 \ -1 \ -1 \ 1 \ -1]^\top$  |
| 6559     | $[-1 \ -1 \ -1 \ -1 \ -1 \ -1 \ -1 \ 1]^\top$  |
| 6560     | $[-1 \ -1 \ -1 \ -1 \ -1 \ -1 \ -1 \ 0]^\top$  |
| 6561     | $[-1 \ -1 \ -1 \ -1 \ -1 \ -1 \ -1 \ -1]^\top$ |

**Table 4** Examples of approximated vectors for the search space  $\mathcal{D}_2$ 

| $n$      | Approximated vector                            |
|----------|------------------------------------------------|
| 1        | $[2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2 \ -1]^\top$        |
| 2        | $[2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 0]^\top$         |
| 3        | $[2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 1]^\top$         |
| 4        | $[2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2]^\top$         |
| $\vdots$ | $\vdots$                                       |
| 390622   | $[-2 \ -2 \ -2 \ -2 \ -2 \ -2 \ -2 \ -2]^\top$ |
| 390623   | $[-2 \ -2 \ -2 \ -2 \ -2 \ -2 \ -2 \ -1]^\top$ |
| 390624   | $[-2 \ -2 \ -2 \ -2 \ -2 \ -2 \ -2 \ 1]^\top$  |
| 390625   | $[-2 \ -2 \ -2 \ -2 \ -2 \ -2 \ -2 \ 0]^\top$  |

subject to constraints such as (i) low-complexity of the candidate vector  $\mathbf{t}$  and (ii) orthogonality of the resulting matrix  $\mathbf{T}$ . The objective function  $\text{error}(\cdot, \cdot)$  returns a given error measure and  $\mathcal{D}$  is a suitable search space.

### 3.2 Search space

In order to obtain a low-complexity matrix  $\mathbf{T}$ , its entries must be computationally simple (Britanak et al. 2007; Blahut 2010). We define the search space as the collection of 8-point vectors whose entries are in a set, say  $\mathcal{P}$ , of low-complexity elements. Therefore, we have the search space  $\mathcal{D} = \mathcal{P}^8$ . Some choices for  $\mathcal{P}$  include:  $\mathcal{P}_1 = \{0, \pm 1\}$  and  $\mathcal{P}_2 = \{0, \pm 1, \pm 2\}$ . Tables 3 and 4 display some elements of the search spaces  $\mathcal{D}_1 = \mathcal{P}_1^8$  and  $\mathcal{D}_2 = \mathcal{P}_2^8$ . These search spaces have 6561 and 390,625 elements, respectively.

### 3.3 Objective function

The problem posed in (5) requires the identification of an error function to quantify the “distance” between the candidate row vectors from  $\mathcal{D}$  and the rows of the exact DCT. Related

literature often consider error functions based on matrix norms (Cintra and Bayer 2011), proximity to orthogonality (Tablada et al. 2015), and coding performance (Britanak et al. 2007).

In this work, we propose the utilization of a distance based on the angle between vectors as the objective function to be minimized. Let  $\mathbf{u}$  and  $\mathbf{v}$  be two vectors defined over the same Euclidean space. The angle between vectors is simply given by:

$$\text{angle}(\mathbf{u}, \mathbf{v}) = \arccos \left( \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \cdot \|\mathbf{v}\|} \right),$$

where  $\langle \cdot, \cdot \rangle$  is the inner product and  $\|\cdot\|$  indicates the norm induced by the inner product (Strang 1988).

### 3.4 Orthogonality and row order

In addition, we require that the ensemble of rows  $\mathbf{t}_k^T, k = 1, 2, \dots, 8$ , must form an orthogonal set. This is to ensure that an orthogonal approximation can be obtained. As shown in Higham (1986) and Cintra (2011), for this property to be satisfied, it suffices that:

$$\mathbf{T} \cdot \mathbf{T}^T = [\text{diagonal matrix}].$$

Because we aim at approximating each of the exact DCT matrix rows individually, the row sequential order according to which the approximations are performed may matter. Notice that we approximate the rows of the DCT based on a set of low-complexity rows, the search space. For instance, let us assume that we approximate the rows in the following order:  $\wp = (1, 2, 3, 4, 5, 6, 7, 8)$ . Once we find a good approximate row for the first exact row, i.e., a row vector in the search space which has the smallest angle in relation to that exact row, the second row is approximated considering only the row vectors in the search space that are orthogonal to the approximation for the first row. After that, the third exact row is approximated considering only the row vectors in the search space that are orthogonal to the first and second rows already chosen. And so on. This procedure characterizes the greedy nature of the proposed algorithm.

Consider now the approximation order  $(4, 3, 5, 6, 1, 2, 7, 8)$ , a permutation of  $\wp$ . In this case, we start by approximating the fourth exact row considering the whole search space because we are starting from it. Hence, the obtained approximate row might be different from the one obtained by considering  $\wp$ , since in that case the search space is restricted in a different manner.

As an example, consider the DCT matrix of length 8, introduced in Sect. 2 of the manuscript. If considering the low complexity set  $\{-1, 0, 1\}$  and the approximation order  $(1, 2, 3, 4, 5, 6, 7, 8)$  we obtain the following approximate matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 \\ 1 & 0 & 0 & -1 & -1 & 0 & 0 & 1 \\ 1 & 0 & -1 & -1 & 1 & 1 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & -1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & -1 & 1 & -1 & 1 & 0 \end{bmatrix}.$$

In other hand, if we consider the reverse approximation order, (8, 7, 6, 5, 4, 3, 2, 1), we obtain the following matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 0 & -1 & -1 & 1 & 1 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 0 & -1 & 1 & -1 & 1 & -1 & 1 & 0 \end{bmatrix}.$$

Therefore, the row sequence considered matters for the resulting matrix. The sequence matters *during* the process of finding the approximate matrix.

Thus, the row vectors  $\mathbf{c}_k^\top$  from the exact matrix must be approximated in all possible sequences. For a systematic procedure, all the  $8! = 40320$  possible permutations of the sequence  $\wp$  must be considered. Let  $\wp_m$ ,  $m = 1, 2, \dots, 40320$ , be the resultant sequence that determines the  $m$ th permutation; e.g.  $\wp_{1250} = (1, 3, 7, 6, 5, 4, 8, 2)$ . The particular elements of a sequence are denoted by  $\wp_m(k)$ ,  $k = 1, 2, \dots, 8$ . Then, for the given example above, we have  $\wp_{1250}(2) = 3$ .

### 3.5 Proposed optimization problem

Considering the above discussion, we can re-cast (5) in more precise terms. For each permutation sequence  $\wp_m$ , we have the following optimization problem:

$$\mathbf{t}_{\wp_m(k)} = \arg \min_{\mathbf{d} \in \mathcal{D}} \text{angle}(\mathbf{c}_{\wp_m(k)}, \mathbf{d}), \quad k = 1, 2, \dots, 8, \quad (6)$$

subject to:

$$\langle \mathbf{t}_{\wp_m(i)}, \mathbf{t}_{\wp_m(j)} \rangle = 0, \quad i \neq j,$$

$m = 1, 2, \dots, 40320$  and a fixed search space  $\mathcal{D} \in \{\mathcal{D}_1, \mathcal{D}_2\}$ . For each  $m$ , the solution of the above problem returns eight vectors,  $\mathbf{t}_{\wp_m(1)}^\top, \mathbf{t}_{\wp_m(2)}^\top, \dots, \mathbf{t}_{\wp_m(8)}^\top$ , that are used as the rows of the desired low-complexity matrix. Note that each sequence  $\wp_m$  may result in a different solution. Effectively, there are  $8! = 40320$  problems to be solved. In principle, each permutation  $\wp_m$  can furnish a different matrix.

Because the search space is relatively small, we solved (6) by means of exhaustive search. Although simple, such approach ensures the attainment of a solution and avoids convergence issues (Cormen et al. 2001). Figure 1 shows the pseudo-code for the adopted procedure to solve (6). It is important to highlight that although the proposed formulation is applicable to arbitrary transform lengths, it may not be computationally feasible. For this reason, we restrict our analysis to the 8-point case. Section 6.2 discusses an alternative form of generating higher order DCT approximations.

## 4 Results and evaluation

In this section, we apply the proposed method aiming at the derivation of new approximations for the 8-point DCT. Subsequently, we analyze and compare the obtained matrices with a



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1: procedure ABMAPPROX( $\mathbf{C}$ ,  $\mathcal{D}$ )
2:   approximations  $\leftarrow$  null 3D matrix of size  $8 \times 8 \times n$ ;
3:   for  $m \leftarrow 1, |\mathcal{D}|$  do
4:      $\mathcal{D}_m \leftarrow \mathcal{D}(m, :)$ 
5:     for  $k \leftarrow 1, 2, \dots, 8$  do
6:        $\theta_{min} \leftarrow 2\pi$ ;
7:        $index \leftarrow 1$ ;
8:       for  $i \leftarrow 1, 2, \dots, |\mathcal{D}|$  do
9:          $aux \leftarrow \mathbf{approximations}(:, :, m) \cdot (\mathcal{D}(i, :))^T$ 
10:        if  $\text{sum}(aux) = 0$  then
11:           $\theta \leftarrow \text{angle}(\mathbf{C}(\mathcal{D}_m(k), :), \mathcal{D}(i, :))$ ;
12:          if  $\theta < \theta_{min}$  then
13:             $\theta_{min} \leftarrow \theta$ ;
14:             $index \leftarrow i$ ;
15:          end if
16:        end if
17:      end for
18:       $\mathbf{approximations}(\mathcal{D}_m(k), :, m) \leftarrow \mathcal{D}(index, :)$ ;
19:    end for
20:  end for
21: end procedure

```

---

**Fig. 1** Algorithm for the proposed method

representative set of DCT approximations described in the literature according to several well-known figures of merit (Salomon et al. 2007).

#### 4.1 New 8-point DCT approximations

Considering the search spaces  $\mathcal{D}_1$  and  $\mathcal{D}_2$  (cf. Tables 3, 4, respectively), we apply the proposed algorithm to solve (6). Because the first and fifth rows of the exact DCT are trivially approximated by the row vectors  $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]$  and  $[1 \ -1 \ -1 \ 1 \ 1 \ -1 \ -1 \ 1]$ , respectively, we limited the search to the remaining six rows. As a consequence, the number of possible candidate matrices is reduced to  $6! = 720$ . For  $\mathcal{D}_1$ , only two different matrices were obtained, which coincide with previously archived approximations, namely: (i) the RDCT (Bayer and Cintra 2010) and (ii) the matrix  $\mathbf{T}_4$  introduced in Cintra et al. (2014). These approximations are depicted in Table 2.

On the other hand, considering the search space  $\mathcal{D}_2$ , the following two new matrices were obtained:

$$\mathbf{T}_1 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 1 & 0 & 0 & -1 & -2 & -2 \\ 2 & 1 & -1 & -2 & -2 & -1 & 1 & 2 \\ 1 & 0 & -2 & -2 & 2 & 2 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 2 & -2 & 0 & 1 & -1 & 0 & 2 & -2 \\ 1 & -2 & 2 & -1 & -1 & 2 & -2 & 1 \\ 0 & -1 & 2 & -2 & 2 & -2 & 1 & 0 \end{bmatrix},$$

$$\mathbf{T}_2 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 2 & 0 & 0 & -2 & -1 & -2 \\ 2 & 1 & -1 & -2 & -2 & -1 & 1 & 2 \\ 2 & 0 & -2 & -1 & 1 & 2 & 0 & -2 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -2 & 0 & 2 & -2 & 0 & 2 & -1 \\ 1 & -2 & 2 & -1 & -1 & 2 & -2 & 1 \\ 0 & -2 & 1 & -2 & 2 & -1 & 2 & 0 \end{bmatrix}.$$

According to (1) and (2), the above *low-complexity* matrices  $\mathbf{T}_1$  and  $\mathbf{T}_2$  can be modified to provide orthogonal transformations  $\hat{\mathbf{C}}_1$  and  $\hat{\mathbf{C}}_2$ . The selected orthogonalization procedure is based on the polar decomposition as described in Higham (1986), Cintra (2011), Higham and Schreiber (1988). Thus, the *orthogonal DCT approximations* associated to the matrices  $\mathbf{T}_1$  and  $\mathbf{T}_2$  are given by

$$\hat{\mathbf{C}}_1 = \mathbf{S}_1 \cdot \mathbf{T}_1 \quad \text{and} \quad \hat{\mathbf{C}}_2 = \mathbf{S}_2 \cdot \mathbf{T}_2,$$

where

$$\mathbf{S}_i = \sqrt{(\mathbf{T}_i \cdot \mathbf{T}_i^T)^{-1}}, \quad i = 1, 2.$$

Thus, it follows that:

$$\mathbf{S}_1 = \mathbf{S}_2 = \text{diag} \left( \frac{1}{\sqrt{8}}, \frac{1}{\sqrt{18}}, \frac{1}{\sqrt{20}}, \frac{1}{\sqrt{18}}, \frac{1}{\sqrt{8}}, \frac{1}{\sqrt{18}}, \frac{1}{\sqrt{20}}, \frac{1}{\sqrt{18}} \right).$$

Other simulations were performed considering extended sets of elements. In particular, following sets were considered:  $\{0, \pm 1, \pm 4\}$ ,  $\{0, \pm 1, \pm 8\}$ ,  $\{0, \pm 1, \pm 2, \pm 4\}$ ,  $\{0, \pm 1, \pm 2, \pm 8\}$ , and  $\{0, \pm 2, \pm 4, \pm 8\}$ . Generally, the resulting matrices did not perform as well as the ones being proposed. Moreover, the associate computational cost was consistently higher.

The number of vectors in the search space can be calculated as  $|\mathcal{D}| = |\mathcal{P}|^8$  (cf. Sect. 3.2). Therefore, including more elements to  $\mathcal{P}$  effects a noticeable increase in the size of the search space. As a consequence, the processing time to derive all the 6! candidate matrices increases accordingly.

## 4.2 Approximation measures

Approximations measurements are computed between an approximate matrix  $\hat{\mathbf{C}}$  (not the low-complexity matrix  $\mathbf{T}$ ) relative to the exact DCT. To evaluate the performance of the proposed approximations,  $\hat{\mathbf{C}}_1$  and  $\hat{\mathbf{C}}_2$ , we selected traditional figures of merit: (i) total error energy ( $\varepsilon(\cdot)$ ) (Cintra and Bayer 2011); (ii) mean square error (MSE( $\cdot$ )) (Britanak et al. 2007; Wang and Bovik 2009); (iii) coding gain ( $C_g(\cdot)$ ) (Britanak et al. 2007; Goyal 2001; Katto and Yasuda 1991); and (iv) transform efficiency ( $\eta(\cdot)$ ) (Britanak et al. 2007). The MSE and total error energy are suitable measures to quantify the difference between the exact DCT and its approximations (Britanak et al. 2007). The coding gain and transform efficiency are appropriate tools to quantify compression, redundancy removal, and data decorrelation capabilities (Britanak et al. 2007). Additionally, due the angular nature of the objective function required by the proposed optimization problem, we also considered descriptive circular statistics (Mardia and Jupp 2009; Jammalamadaka and Sengupta 2001). Circular statistics allows the quantification of approximation error in terms of the angle difference between the row vectors of the approximated and exact matrix.

Hereafter we adopt the following quantities and notation: the interpixel correlation is  $\rho = 0.95$  (Britanak et al. 2007; Goyal 2001; Liang and Tran 2001),  $\widehat{\mathbf{C}}$  is an approximation for the DCT, and  $\widehat{\mathbf{R}}_{\mathbf{y}} = \widehat{\mathbf{C}} \cdot \mathbf{R}_{\mathbf{x}} \cdot \widehat{\mathbf{C}}^{\top}$ , where  $\mathbf{R}_{\mathbf{x}}$  is the covariance matrix of  $\mathbf{x}$ , whose elements are given by  $\rho^{|i-j|}$ ,  $i, j = 1, 2, \dots, 8$ . We detail each of these measures below.

#### 4.2.1 Total energy error

The total energy error is a similarity measure given by (Cintra and Bayer 2011):

$$\varepsilon(\widehat{\mathbf{C}}) = \pi \cdot \|\mathbf{C} - \widehat{\mathbf{C}}\|_{\text{F}}^2,$$

where  $\|\cdot\|_{\text{F}}$  represents the Frobenius norm (Watkins 2004).

#### 4.2.2 Mean square error

The MSE of a given approximation  $\widehat{\mathbf{C}}$  is furnished by (Britanak et al. 2007; Wang and Bovik 2009):

$$\text{MSE}(\widehat{\mathbf{C}}) = \frac{1}{8} \cdot \text{tr} \left( (\mathbf{C} - \widehat{\mathbf{C}}) \cdot \mathbf{R}_{\mathbf{x}} \cdot (\mathbf{C} - \widehat{\mathbf{C}})^{\top} \right),$$

where  $\text{tr}(\cdot)$  represents the trace operator (Britanak et al. 2007). The total energy error and the mean square error are appropriate measures for capturing the approximation error in a Euclidean distance sense.

#### 4.2.3 Coding gain

The coding gain quantifies the energy compaction capability and is given by (Britanak et al. 2007):

$$C_g(\widehat{\mathbf{C}}) = 10 \cdot \log_{10} \left\{ \frac{\frac{1}{8} \sum_{i=1}^8 r_{i,i}^2}{\left( \prod_{i=1}^8 r_{i,i}^2 \cdot \|\widehat{\mathbf{c}}_i\|^2 \right)^{1/8}} \right\},$$

where  $r_{i,i}$  is the  $i$ th element of the diagonal of  $\widehat{\mathbf{R}}_{\mathbf{y}}$  (Britanak et al. 2007) and  $\widehat{\mathbf{c}}_i^{\top}$  is the  $i$ th row of  $\widehat{\mathbf{C}}$ .

However, as pointed in Katto and Yasuda (1991), the previous definition is suitable for orthogonal transforms only. For non-orthogonal transforms, such as SDCT (Haweel 2001) and MRDCT (Bayer and Cintra 2012), we adopt the unified coding gain (Katto and Yasuda 1991). For  $i = 1, 2, \dots, 8$ , let  $\widehat{\mathbf{c}}_i^{\top}$  and  $\widehat{\mathbf{g}}_i^{\top}$  be  $i$ th row of  $\widehat{\mathbf{C}}$  and  $\widehat{\mathbf{C}}^{-1}$ , respectively. Then, the unified coding gain is given by:

$$C_g^*(\widehat{\mathbf{C}}) = 10 \cdot \log_{10} \left\{ \prod_{i=1}^8 \frac{1}{\sqrt[8]{A_i \cdot B_i}} \right\},$$

where  $A_i = \text{su}[(\widehat{\mathbf{c}}_i \cdot \widehat{\mathbf{c}}_i^{\top}) \odot \mathbf{R}_{\mathbf{x}}]$ ,  $\text{su}(\cdot)$  returns the sum of all elements of the input matrix, the operator  $\odot$  represents the element-wise product, and  $B_i = \|\widehat{\mathbf{g}}_i\|^2$ .

#### 4.2.4 Transform efficiency

The transform efficiency is an alternative measure to the coding gain, being expressed according to (Britanak et al. 2007):

$$\eta(\hat{\mathbf{C}}) = \frac{\sum_{i=1}^8 |r_{i,i}|}{\sum_{i=1}^8 \sum_{j=1}^8 |r_{i,j}|} \cdot 100,$$

where  $r_{i,j}$  is the  $(i, j)$ th entry of  $\hat{\mathbf{R}}_{\mathbf{y}}$ ,  $i, j = 1, 2, \dots, 8$ .

#### 4.2.5 Circular statistics

Because the objective function in (6) is the operator angle, its associate values are distributed around the unit circle. This type of data is suitably analyzed by circular statistics tools (Mardia and Jupp 2009; Jammalamadaka and Sengupta 2001; Mahan 1991). Let  $\mathbf{a}$  be an arbitrary 8-point vector and  $\mathbf{q} = [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$ . The angle function is given by (Mardia and Jupp 2009):

$$\theta = \text{angle}(\mathbf{a}', \mathbf{q}), \quad k = 1, 2, \dots, 8,$$

where  $\mathbf{a}' = \frac{\mathbf{a}}{\|\mathbf{a}\|}$  is the normalized vector of  $\mathbf{a}$ .

The mean angle (circular mean) is given by (Mahan 1991; Jammalamadaka and Sengupta 2001):

$$\bar{\theta} = \begin{cases} \arctan(S/C), & \text{if } C > 0 \text{ and } S \geq 0, \\ \pi/2, & \text{if } C = 0 \text{ and } S > 0, \\ \arctan(S/C) + \pi, & \text{if } C < 0, \\ \arctan(S/C) + 2\pi, & \text{if } C \geq 0 \text{ and } S < 0, \\ \text{undefined}, & \text{if } C = 0 \text{ and } S = 0, \end{cases}$$

where  $C = \sum_i \cos(\theta_i)$ ,  $S = \sum_i \sin(\theta_i)$ , and  $\{\theta_i\}$  is a collection of angles. The circular variance is given by (Mardia and Jupp 2009):

$$V = 1 - \frac{\sqrt{C^2 + S^2}}{8}.$$

The minimal variance occurs when all observed angles are identical. In this case, we have  $V = 0$ . In other hand, the maximum variance occurs when the observations are uniformly distributed around the unit circle. Thus,  $V = 1$  (Jammalamadaka and Sengupta 2001).

Considering the rows of the 8-point DCT matrix and of a given 8-point DCT approximate matrix, the angle function furnishes the following angles, respectively:  $\theta_{\mathbf{c}_k} = \text{angle}(\mathbf{c}_k, \mathbf{q})$  and  $\theta_{\mathbf{t}_k} = \text{angle}(\mathbf{t}_k, \mathbf{q})$ ,  $k = 1, 2, \dots, 8$  [cf. (3) and (4)]. In this particular case, the mean circular difference, which measures the mean difference between the pairs of angles is defined as follows:

$$\bar{D} = \frac{1}{8^2} \cdot \sum_{i=1}^8 \sum_{j=1}^8 (\pi - |\pi - |\theta_{\mathbf{c}_i} - \theta_{\mathbf{t}_j}||).$$

The expression above considers the difference between all the possible pairs of angles. However, we are interested in comparing the angle between the  $i$ th row of the DCT and

**Table 5** Performance measures for DCT approximations derived from low-complexity matrices

| Method                                                         | $\varepsilon$ | MSE                   | $C_g^*$ | $\eta$  |
|----------------------------------------------------------------|---------------|-----------------------|---------|---------|
| DCT (Ahmed et al. 1974)                                        | 0             | 0                     | 8.8259  | 93.9912 |
| IDCT (HEVC) (Ohm et al. 2012)                                  | 0.0020        | $8.66 \times 10^{-6}$ | 8.8248  | 93.8236 |
| $\hat{\mathbf{C}}_1$ (proposed)                                | 1.2194        | 0.0046                | 8.6337  | 90.4615 |
| $\hat{\mathbf{C}}_2$ (proposed)                                | 1.2194        | 0.0127                | 8.1024  | 87.2275 |
| $\hat{\mathbf{C}}_{\text{LO}}$ (Lengwehasatit and Ortega 2004) | 0.8695        | 0.0061                | 8.3902  | 88.7023 |
| $\hat{\mathbf{C}}_{\text{SDCT}}$ (Haweel 2001)                 | 3.3158        | 0.0207                | 6.0261  | 82.6190 |
| $\hat{\mathbf{C}}_{\text{RDCT}}$ (Bayer and Cintra 2010)       | 1.7945        | 0.0098                | 8.1827  | 87.4297 |
| $\hat{\mathbf{C}}_{\text{MRDCT}}$ (Bayer and Cintra 2012)      | 8.6592        | 0.0594                | 7.3326  | 80.8969 |
| $\hat{\mathbf{C}}_{\text{BAS-2008a}}$ (Bouguezet et al. 2008)  | 5.9294        | 0.0238                | 8.1194  | 86.8626 |
| $\hat{\mathbf{C}}_{\text{BAS-2008b}}$ (Bouguezet et al. 2008)  | 4.1875        | 0.0191                | 6.2684  | 83.1734 |
| $\hat{\mathbf{C}}_{\text{BAS-2009}}$ (Bouguezet et al. 2009)   | 6.8543        | 0.0275                | 7.9126  | 85.3799 |
| $\hat{\mathbf{C}}_{\text{BAS-2011}}$ (Bouguezet et al. 2011)   | 26.8462       | 0.0710                | 7.9118  | 85.6419 |
| $\hat{\mathbf{C}}_{\text{BAS-2013}}$ (Bouguezet et al. 2013)   | 35.0639       | 0.1023                | 7.9461  | 85.3138 |
| $\hat{\mathbf{C}}'_1$ (Cintra et al. 2014)                     | 3.3158        | 0.0208                | 6.0462  | 83.0814 |
| $\hat{\mathbf{C}}_4$ (Cintra et al. 2014)                      | 1.7945        | 0.0098                | 8.1834  | 87.1567 |
| $\hat{\mathbf{C}}_5$ (Cintra et al. 2014)                      | 1.7945        | 0.0100                | 8.1369  | 86.5359 |
| $\hat{\mathbf{C}}_6$ (Cintra et al. 2014)                      | 0.8695        | 0.0062                | 8.3437  | 88.0594 |

Exact DCT measures listed for reference

the corresponding row of the approximated matrix, i.e., the cases where  $i = j$ . Thus we have the modified circular mean difference according to:

$$\bar{D}_{mod} = \frac{1}{8} \cdot \sum_{i=1}^8 (\pi - |\pi - |\theta_{c_i} - \theta_{t_i}||).$$

### 4.3 Results and comparisons

Table 5 shows the obtained measurements for all approximations derived, according to (1), from the low-complexity matrices considered in this paper. Table 6 brings a summary of the descriptive circular statistics. We also included the exact DCT and the integer DCT (IDCT) (Ohm et al. 2012) for comparison. The considered IDCT is the 8-point approximation adopted in the HEVC standard (Ohm et al. 2012). A more detailed analysis on the performance of the proposed approximation in comparison with the IDCT is provided in Sect. 6.2. The proposed DCT approximation  $\hat{\mathbf{C}}_1$  outperforms all competing methods in terms of MSE, coding gain, and transform efficiency. It also performs as the second best for total error energy measurement. It is unusual for an approximation to simultaneously excel in measures based on Euclidean distance ( $\varepsilon$  and MSE) as well as in coding-based measures. The approximation by Lengwehasatit–Ortega ( $\hat{\mathbf{C}}_{\text{LO}}$ ) (Lengwehasatit and Ortega 2004) achieves second best results MSE, and  $\eta$ . Because of its relatively inferior performance, we removed the new approximation  $\hat{\mathbf{C}}_2$  from our subsequent analysis. Nevertheless,  $\hat{\mathbf{C}}_2$  could outperform the approximations  $\hat{\mathbf{C}}_{\text{BAS-2008b}}$  (Bouguezet et al. 2008),  $\hat{\mathbf{C}}_{\text{BAS-2009}}$  (Bouguezet et al. 2009),  $\hat{\mathbf{C}}_{\text{BAS-2011}}$  (Bouguezet et al. 2011),  $\hat{\mathbf{C}}_{\text{BAS-2013}}$  (Bouguezet et al. 2013),  $\hat{\mathbf{C}}_{\text{SDCT}}$  (Haweel 2001),  $\hat{\mathbf{C}}_{\text{MRDCT}}$  (Bayer and

**Table 6** Descriptive circular statistics

| Method                                   | $\bar{\theta}$ | $V$    | $\bar{D}_{mod}$ |
|------------------------------------------|----------------|--------|-----------------|
| Exact DCT (Ahmed et al. 1974)            | 70.53          | 0.0089 | 0               |
| IDCT (HEVC) (Ohm et al. 2012)            | 70.50          | 0.0086 | 0.0022          |
| $T_1$ (proposed)                         | 71.12          | 0.0124 | 0.0711          |
| $T_2$ (proposed)                         | 71.12          | 0.0124 | 0.0343          |
| $T_{LO}$ (Lengwehasatit and Ortega 2004) | 70.81          | 0.0102 | 0.0483          |
| $T_{SDCT}$ (Haweel 2001)                 | 69.29          | 0      | 0.1062          |
| $T_{RDCT}$ (Bayer and Cintra 2010)       | 71.98          | 0.0174 | 0.0716          |
| $T_{MRDCT}$ (Bayer and Cintra 2012)      | 75.58          | 0.0392 | 0.1646          |
| $T_{BAS-2008a}$ (Bouguezel et al. 2008)  | 72.35          | 0.0198 | 0.1036          |
| $T_{BAS-2008b}$ (Bouguezel et al. 2008)  | 67.29          | 0.0015 | 0.1097          |
| $T_{BAS-2009}$ (Bouguezel et al. 2009)   | 72.10          | 0.0183 | 0.1334          |
| $T_{BAS-2011}$ (Bouguezel et al. 2011)   | 73.54          | 0.0265 | 0.1492          |
| $T_{BAS-2013}$ (Bouguezel et al. 2013)   | 69.29          | 0      | 0.1062          |
| $T'_1$ (Cintra et al. 2014)              | 73.54          | 0.0265 | 0.0901          |
| $T_4$ (Cintra et al. 2014)               | 70.57          | 0.0085 | 0.0781          |
| $T_5$ (Cintra et al. 2014)               | 72.45          | 0.0209 | 0.0730          |
| $T_6$ (Cintra et al. 2014)               | 71.27          | 0.0139 | 0.0497          |

Cintra 2012), and  $\hat{C}'_1$  (Cintra et al. 2014) in all measures considered,  $\hat{C}_4$  (Cintra et al. 2014) and  $\hat{C}_5$  (Cintra et al. 2014) in terms of total error energy and transform efficiency,  $\hat{C}_{RDCT}$  (Bayer and Cintra 2010) in terms of total error energy, and  $\hat{C}_{BAS-2008a}$  (Bouguezel et al. 2008) in terms of total error energy, MSE and transform efficiency. Hereafter we focus our attention on the proposed approximation  $\hat{C}_1$ .

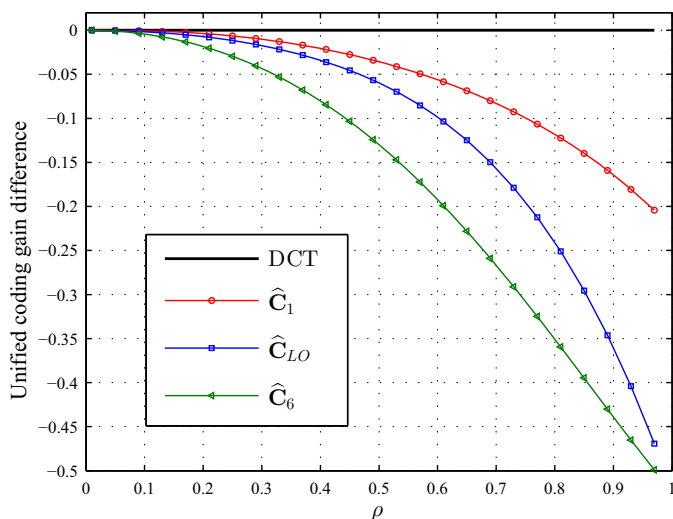
The proposed search algorithm is greedy, i.e., it makes local optimal choices hoping to find the global optimum solution (Cormen et al. 2001). Therefore, it is not guaranteed that the obtained solutions are globally optimal. This is exactly what happens here. As can be seen in Table 6, the proposed matrix  $T_1$  is not the transformation matrix that provides the lowest circular mean difference among the approximations on literature. Despite this fact, the proposed matrix has outstanding performance according to the considered measures.

Figure 2 shows the effect of the interpixel correlation  $\rho$  on the performance of the discussed approximate transforms as measured by the unified coding gain difference compared to the exact DCT (Han et al. 2013). The proposed method outperforms the competing methods as its coding gain difference is smaller for any choice of  $\rho$ . For highly correlated data the coding degradation in dB is roughly reduced by half when the proposed approximation  $\hat{C}_1$  is considered.

## 5 Fast algorithm and hardware realization

### 5.1 Fast algorithm

The direct implementation of  $T_1$  requires 48 additions and 24 bit-shifting operations. However, such computational cost can be significantly reduced by means of sparse matrix



**Fig. 2** Curves for the coding gain error of  $\hat{C}_1$ ,  $\hat{C}_{LO}$ , and  $\hat{C}_6$ , relative to the exact DCT, for  $0 < \rho < 1$

factorization (Blahut 2010). In fact, considering butterfly-based structures as commonly found in decimation-in-frequency algorithms, such as Hou (1987), Yip and Rao (1988), Rao and Yip (1990), we could derive the following factorization for  $T_1$ :

$$T_1 = D \cdot A_4 \cdot A_3 \cdot A_2 \cdot A_1,$$

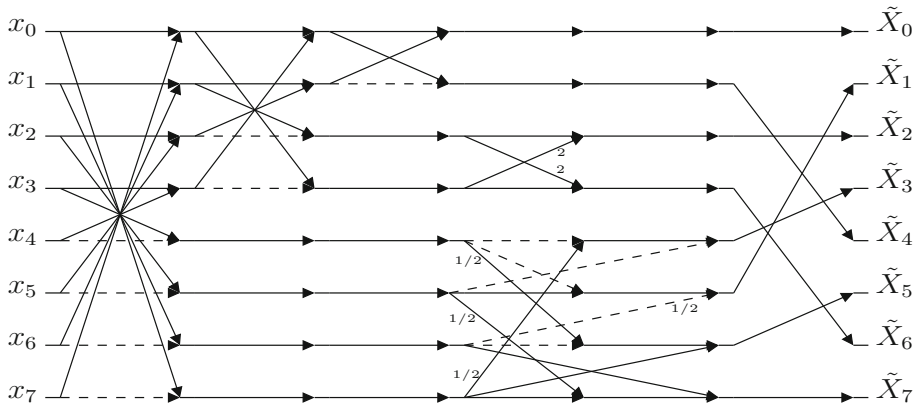
where:

$$A_1 = \begin{bmatrix} 1 & & & & & & & 1 \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ 1 & & & & & & & -1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & & & & & & & 1 \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & -1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix},$$

$$A_3 = \begin{bmatrix} 1 & & & & & & & \\ 1 & & & & & & & \\ & -1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 2 & & & & & \\ & & & -1 & & & & \\ & & & & -1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix},$$

and  $D = \text{diag}(1, 2, 1, 2, 1, 2, 1, 2)$ . Figure 3 shows the signal flow graph (SFG) related to the above factorization. The computational cost of this algorithm is only 24 additions and six multiplications by two. The multiplications by two are extremely simple to be performed, requiring only bit-shifting operations (Britanak et al. 2007). The fast algorithm proposed requires 50% less additions and 75% less bit-shifting operations when compared to the direct implementation. The computational costs of the considered methods are shown in Table 7. The additive cost of the discussed approximations varies from 14 to 28 additions.

In general terms, DCT approximations exhibit a trade-off between computational cost and transform performance (Tablada et al. 2015), i.e., less complex matrices effect poor spectral approximations (Britanak et al. 2007). Departing from this general behavior, the



**Fig. 3** Signal flow graph of the proposed transform, relating the input data  $x_n$ ,  $n = 0, 1, \dots, 7$ , to its correspondent coefficients  $\tilde{X}_k$ ,  $k = 0, 1, \dots, 7$ , where  $\tilde{\mathbf{X}} = \mathbf{T}_1 \cdot \mathbf{x}$ . Dashed arrows representing multiplication by  $-1$

**Table 7** Computational cost comparison

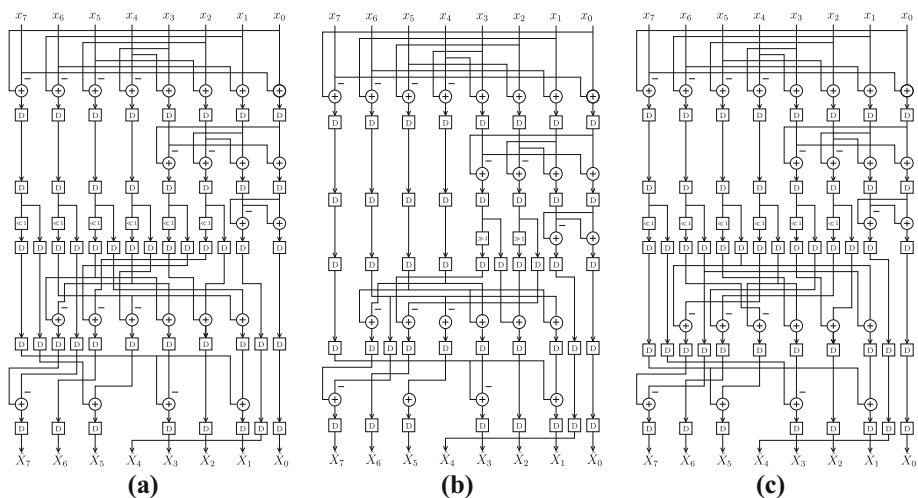
| Method                                            | Multiplications | Additions | Bit-shifts |
|---------------------------------------------------|-----------------|-----------|------------|
| DCT (Loeffler et al. 1989)                        | 11              | 29        | 0          |
| IDCT (HEVC) (Ohm et al. 2012)                     | 0               | 50        | 30         |
| $\mathbf{T}_1$ (proposed)                         | 0               | 24        | 6          |
| $\mathbf{T}_{LO}$ (Lengwehasatit and Ortega 2004) | 0               | 24        | 2          |
| $\mathbf{T}_{SDCT}$ (Haweel 2001)                 | 0               | 24        | 0          |
| $\mathbf{T}_{RDCT}$ (Bayer and Cintra 2010)       | 0               | 22        | 0          |
| $\mathbf{T}_{MRDCT}$ (Bayer and Cintra 2012)      | 0               | 14        | 0          |
| $\mathbf{T}_{BAS-2008a}$ (Bouguezet et al. 2008)  | 0               | 18        | 2          |
| $\mathbf{T}_{BAS-2008b}$ (Bouguezet et al. 2008)  | 0               | 21        | 0          |
| $\mathbf{T}_{BAS-2009}$ (Bouguezet et al. 2009)   | 0               | 18        | 0          |
| $\mathbf{T}_{BAS-2011}$ (Bouguezet et al. 2011)   | 0               | 16        | 0          |
| $\mathbf{T}_{BAS-2013}$ (Bouguezet et al. 2013)   | 0               | 24        | 0          |
| $\mathbf{T}'_1$ (Cintra et al. 2014)              | 0               | 18        | 0          |
| $\mathbf{T}_4$ (Cintra et al. 2014)               | 0               | 24        | 0          |
| $\mathbf{T}_5$ (Cintra et al. 2014)               | 0               | 24        | 4          |
| $\mathbf{T}_6$ (Cintra et al. 2014)               | 0               | 24        | 6          |

proposed transformation  $\mathbf{T}_1$  has (i) excelling performance measures and (ii) lower or similar arithmetic cost when compared to competing methods, as shown in Tables 5, 6, and 7. Regarding considered performance measures, three transformations are consistently placed among the five best methods:  $\mathbf{T}_1$ ,  $\mathbf{T}_{LO}$ , and  $\mathbf{T}_6$ . Thus, we separate them for hardware analysis.

## 5.2 FPGA implementation

The proposed design along with  $\mathbf{T}_{LO}$  and  $\mathbf{T}_6$  were implemented on an FPGA chip using the Xilinx ML605 board. Considering hardware co-simulation the FPGA realization was tested with 100,000 random 8-point input test vectors. The test vectors were generated from within





**Fig. 4** Architectures for **a**  $T_1$ , **b**  $T_{LO}$ , **c**  $T_6$

**Table 8** Hardware resource consumption and power consumption using Xilinx Virtex-6 XC6VLX240T 1FFG1156 device

| Approximation                            | CLB | FF  | $T_{cpd}$ (ns) | $F_{max}$ (MHz) | $D_p$ (mW/GHz) | $Q_p$ (W) | $AT$ | $AT^2$ |
|------------------------------------------|-----|-----|----------------|-----------------|----------------|-----------|------|--------|
| $T_1$ (proposed)                         | 135 | 408 | 1.750          | 571             | 2.74           | 3.471     | 236  | 413    |
| $T_{LO}$ (Lengwehasatit and Ortega 2004) | 114 | 349 | 1.900          | 526             | 2.82           | 3.468     | 217  | 412    |
| $T_6$ (Cintra et al. 2014)               | 125 | 389 | 2.100          | 476             | 2.57           | 3.460     | 262  | 551    |

the MATLAB environment and, using JTAG based hardware co-simulation, routed to the physical FPGA device where each algorithm was realized in the reconfigurable logic fabric. Then the computational results obtained from the FPGA algorithm implementations were routed back to MATLAB memory space. The diagrams for the designs can be seen in Fig. 4.

The metrics employed to evaluate the FPGA implementations were: configurable logic blocks (CLB), flip-flop (FF) count, and critical path delay ( $T_{cpd}$ ), in ns. The maximum operating frequency was determined by the critical path delay as  $F_{max} = (T_{cpd})^{-1}$ , in MHz. Values were obtained from the Xilinx FPGA synthesis and place-route tools by accessing the `xflow.results` report file. Using the Xilinx XPower Analyzer, we estimated the static ( $Q_p$  in W) and dynamic power ( $D_p$  in mW/MHz) consumption. In addition, we calculated area-time ( $AT$ ) and area-time-square ( $AT^2$ ) figures of merit, where area is measured as the CLBs and time as the critical path delay. The values of those metrics for each design are shown in Table 8.

The design linked to the proposed design approximation  $T_1$  possesses the smallest  $T_{cpd}$  among the considered methods. Such critical path delay allows for operations at a 8.55% and 19.96% higher frequency than the designs associated to  $T_{LO}$  and  $T_6$ , respectively. In terms of area-time and are-time-square measures, the design linked to the approximation  $T_{LO}$  presents the best results, followed by the one associated to  $T_1$ .

## 6 Computational experiments

### 6.1 Still image compression

#### 6.1.1 Experiment setup and results

To evaluate the efficiency of the proposed transformation matrix, we performed a JPEG-like image compression experiments as described in Cintra et al. (2014), Potluri et al. (2014), Bayer and Cintra (2010). Input images were divided into sub-blocks of size  $8 \times 8$  pixels and submitted to a bidimensional (2-D) transformation, such as the DCT or one of its approximations. Let  $\mathbf{A}$  be a sub-block of size  $8 \times 8$ . The 2-D approximate transform of  $\mathbf{A}$  is an  $8 \times 8$  sub-block  $\mathbf{B}$  obtained as follows (Cintra et al. 2014; Cintra and Bayer 2011):

$$\mathbf{B} = \hat{\mathbf{C}} \cdot \mathbf{A} \cdot \hat{\mathbf{C}}^T.$$

Considering the zig-zag scan pattern as detailed in Pao and Sun (1998), the initial  $r$  elements of  $\mathbf{B}$  were retained; whereas the remaining  $(64 - r)$  elements were discarded. Considering 8-bit images, this approach implies that the fixed average bits per pixel equals  $r/8$  bits per pixel (bpp). The previous operation results in a matrix  $\mathbf{B}'$  populated with zeros which is suitable for entropy encoding (Wallace 1992). Each processed sub-block was submitted to the corresponding 2-D inverse transformation and the image was reconstructed. For orthogonal approximations, the 2-D inverse transform is given by:

$$\mathbf{A} = \hat{\mathbf{C}}^T \cdot \mathbf{B}' \cdot \hat{\mathbf{C}}.$$

We considered 44 standardized images obtained from the ‘miscellaneous’ volume from USC-SIPI image bank (2017), which include common images such as *Lena*, *Boat*, *Baboon*, and *Peppers*. Without loss of generality, images were converted to 8-bit grayscale and submitted to the above described procedure. The reconstructed images were compared with the original images and evaluated quantitatively according to popular figures of merit: the mean square error (MSE) (Britanak et al. 2007), the peak signal-to-noise ratio (PSNR) (Salomon et al. 2007) and the structural similarity index (SSIM) (Wang et al. 2004). We consider the MSE and PSNR measures because of its good properties and historical usage. However, as discussed in Wang and Bovik (2009), the MSE and PSNR are not the best measures when it comes to predict human perception of image fidelity and quality, for which SSIM has been shown to be a better measure (Wang et al. 2004; Wang and Bovik 2009).

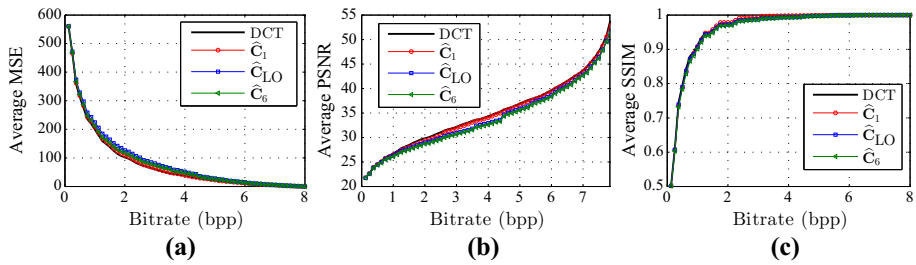
Figure 5 shows the average MSE, PSNR, and SSIM respectively, for the 44 images considering  $1 < r < 64$  (bpp from 0 to 8) retained coefficients. The proposed approximation  $\hat{\mathbf{C}}_1$  outperforms  $\hat{\mathbf{C}}_{LO}$  and  $\hat{\mathbf{C}}_6$  in terms of MSE and PSNR for any value of  $r$ . In terms of SSIM,  $\hat{\mathbf{C}}_1$  outperforms  $\hat{\mathbf{C}}_6$  for any value of  $r$  and  $\hat{\mathbf{C}}_{LO}$  for  $r \in [7, 63]$ .

In order to better visualize previous curves, we adopted the relative difference which is given by (Higham 2008):

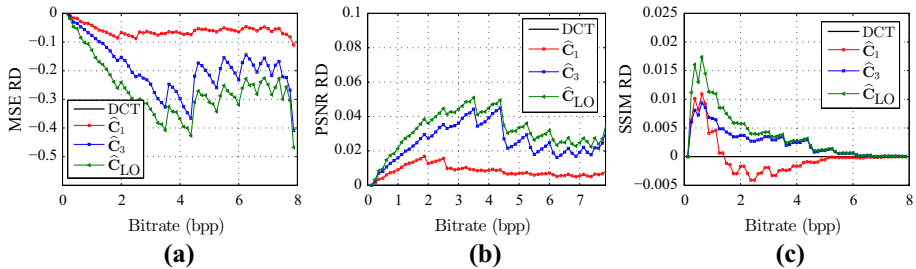
$$RD = \frac{\mu(\mathbf{C}) - \mu(\hat{\mathbf{C}})}{\mu(\mathbf{C})},$$

where  $\mu(\mathbf{C})$  and  $\mu(\hat{\mathbf{C}})$  indicate measurements according to the exact and approximate DCT, respectively; and  $\mu \in \{\text{MSE}, \text{PSNR}, \text{SSIM}\}$ .

The relative difference for the MSE, PSNR, and SSIM are presented in Fig. 6. Figure 6c shows that, for  $12 < r < 60$  (bpp from 1.5 to 7.5),  $\hat{\mathbf{C}}_1$  outperforms not only  $\hat{\mathbf{C}}_{LO}$  and  $\hat{\mathbf{C}}_6$  but the DCT itself. To the best of our knowledge, this particularly good behavior was



**Fig. 5** Curves for the average of **a** MSE, **b** PSNR, **c** SSIM corresponding to 44 images



**Fig. 6** Relative difference curves for **a** MSE, **b** PSNR, **c** SSIM of  $\hat{C}_1$ ,  $\hat{C}_{LO}$ , and  $\hat{C}_6$ , relative to the exact DCT

never described in literature, where invariably the performance of DCT approximations are routinely bounded by the performance of the exact DCT.

A qualitative evaluation is provided in Figs. 7 and 8, where the reconstructed *Lena* images (2017) for  $r = 3$  (0.325 bpp) and  $r = 14$  (1.75 bpp), respectively, according to the exact DCT,  $\hat{C}_1$ ,  $\hat{C}_{LO}$ , and  $\hat{C}_6$  are shown. As expected from the results shown in Fig. 6c, for a bitrate lower than 1.5, the proposed approximate transform matrix is not the one that performs the best (Fig. 7), although the results are very similar to the ones furnished by the exact DCT. For a bitrate value larger than 1.5, Fig. 8 demonstrates a situation where the proposed approximation overcomes the other transforms, including the DCT. In both cases, the visual difference between the DCT and the proposed approximate transform matrix is very small.

### 6.1.2 Discussion

The obtained approximation was capable of outperforming the DCT under the above described conditions. We think that this is relevant, because it directly offers a counter-example to the belief that the coding performance of an approximation is supposed to always be inferior to the DCT. The theoretical background that leads the optimal performance of the DCT is based on assumption that the considered images must follow the Markov-1 processes with high correlation ( $\rho \rightarrow 1$ ). In practice, natural images tend to fit under this assumption, but at lower correlation values the ideal case ( $\rho \rightarrow 1$ ) may not necessarily be met as strongly. For instance, the average correlation of the considered image set was roughly 0.86. This practical deviation from the ideal case ( $\rho \rightarrow 1$ ) may also play a role in explaining our results.

Finding matrix approximation as described in this work is a computational and numerical task. To the best of our knowledge, we cannot identify any methodology that could furnish necessary mathematical tools to design optimal approximations for image compression in



**Fig. 7** Compression of Lena using **a** DCT, **b**  $\hat{C}_1$ , **c**  $\hat{C}_{LO}$ , **d**  $\hat{C}_6$  considering  $r = 3$  (0.325 bpp). **a** MSE = 119.91, PSNR = 27.34, SSIM = 0.8814, **b** MSE = 124.44, PSNR = 27.18, SSIM = 0.8767, **c** MSE = 131.08, PSNR = 26.95, SSIM = 0.8781, **d** MSE = 129.03, PSNR = 27.02, SSIM = 0.8763

an *a priori* manner, i.e., before search space approaches, optimization problem solving, or numerical simulation. In Britanak et al. (2007), Rao and Yip (1990), Tablada et al. (2015), Coutinho et al. (2015), Chan and Lee (2006) and Dimitrov et al. (1998), a large number of methods is listed; all of them aim at good approximations. Although the problem of matrix approximation is quite simple to state, it is also very tricky and offers several non-linearities when combined to a more sophisticated system, such as image compression codecs. Finding low-complexity matrices can be categorized as an integer optimization problem. Thus, navigating in the low-complexity matrix search space might generate non-trivial performance curves, usually leading to discontinuities, which seems to be the case of the proposed approximations matrix  $T_1$ . The navigation path through the search space is highly dependent on the search method and its objective function. Although approximation methods are very likely to provide reasonable approximations, it is also very hard to tell beforehand whether a given approximation method is capable of furnishing extremely good results capable of outperforming the state of the art. Only after experimentation with the obtained approximations one may know better. In particular, this work advances an optimization setup based on a geo-



**Fig. 8** Compression of Lena using **a** DCT, **b**  $\hat{\mathbf{C}}_1$ , **c**  $\hat{\mathbf{C}}_{LO}$ , **d**  $\hat{\mathbf{C}}_6$  considering  $r = 14$  (1.75). **a** MSE = 27.17, PSNR = 33.78, SSIM = 0.9888, **b** MSE = 33.48, PSNR = 32.88, SSIM = 0.9893, **c** MSE = 40.07, PSNR = 32.10, SSIM = 0.9849, **d** MSE = 42.18, PSNR = 31.87, SSIM = 0.9844.

metrically intuitive objective function (angle between vectors) that could find good matrices as demonstrated by our *a posteriori* experiments.

## 6.2 Video coding

In order to assess the proposed transform  $\hat{\mathbf{C}}_1$  as a tool for video coding, we embedded it into a public available (HEVC reference software 2013). The HEVC presents several improvements relative to its predecessors (Sullivan et al. 2012) and aims at providing high compression rates (Pourazad et al. 2012). Differently from other standards (cf. Sect. 1), HEVC employs not only an 8-point integer DCT (IDCT) but also transforms of size 4, 16, and 32 (Ohm et al. 2012). Such feature effects a series of optimization routines allowing the processing of big smooth or textureless areas (Pourazad et al. 2012).

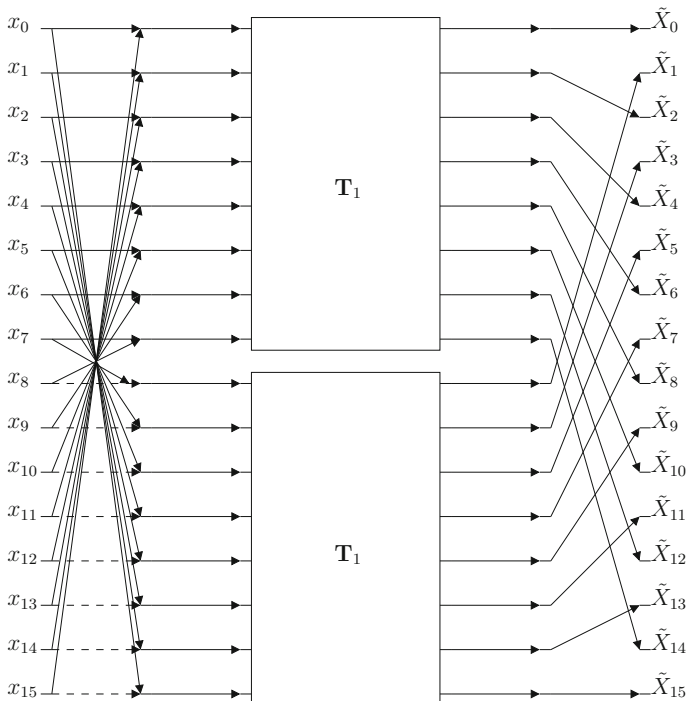
The computational search used in the proposed method proved feasible for the 8-point case. However, as  $N$  increases, the size of the search space increases very quickly. For the case  $N = 16$ , if considering the low complexity set  $\mathcal{P} = \{-1, 0, 1\}$ , we have a search space with  $3^{16} \approx 4.3 \times 10^7$  elements. For this blocklength, the proposed algorithm takes

about 6 minutes to find an approximation for a fixed row sequence, considering a machine with the following specifications: 16-core 2.4 GHZ Intel(R) Xeon(R) CPU E5-2630 v3, with 32 GB RAM running Ubuntu 16.04.3 LTS 64-bit. Therefore, since we have  $N!$  matrices to be generated, to run the whole algorithm would take approximately  $16! \times 6$  minutes. In other words, the computation time would take an extremely large amount of time. Thus, for now, we limited the scope of our computational search to 8-point approximations.

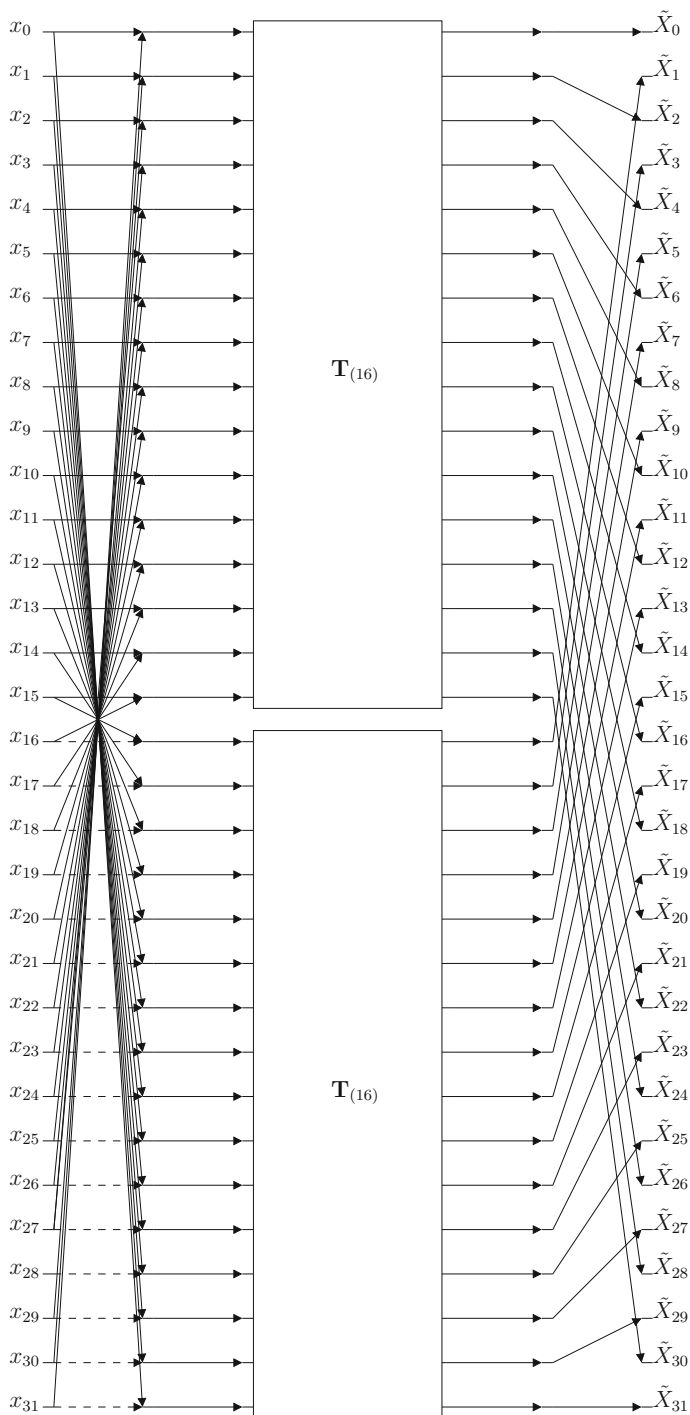
For this reason, aiming to derive large blocklength transforms for HEVC embedding, we submitted the proposed transform matrix  $T_1$  to the Jridi–Alfalou–Meher (JAM) scalable algorithm (Jridi et al. 2015). Such method resulted in 16- and 32-point versions of the proposed matrix  $T_1$  that are suitable for the sought video experiments. Although the JAM algorithm is similar to Chen’s DCT (Chen et al. 1977), it exploits redundancies allowing concise and high parallelizable hardware implementations (Jridi et al. 2015). From a low-complexity  $N/2$ -point transform, the JAM algorithm generates an  $N \times N$  matrix transformation by combining two instantiations of the smaller one. The larger  $N$ -point transform is recursively defined by:

$$T_{(N)} = \frac{1}{\sqrt{2}} M_N^{\text{per}} \begin{bmatrix} T_{\left(\frac{N}{2}\right)} & Z_{\frac{N}{2}} \\ Z_{\frac{N}{2}} & T_{\left(\frac{N}{2}\right)} \end{bmatrix} M_N^{\text{add}}, \quad (7)$$

where  $Z_{\frac{N}{2}}$  is a matrix of order  $N/2$  with all zeroed entries. Matrices  $M_N^{\text{add}}$  and  $M_N^{\text{per}}$  are, respectively, obtained according to:



**Fig. 9** SFG for the proposed 16-point low complexity transform matrix



**Fig. 10** SFG for the proposed 32-point low complexity transform matrix, where  $T_{(16)}$  is the 16-point matrix presented in Fig. 9



$$\mathbf{M}_N^{\text{add}} = \begin{bmatrix} \mathbf{I}_{N/2} & \bar{\mathbf{I}}_{N/2} \\ \bar{\mathbf{I}}_{N/2} & -\mathbf{I}_{N/2} \end{bmatrix}$$

and

$$\mathbf{M}_N^{\text{per}} = \begin{bmatrix} \mathbf{P}_{N-1, N/2} & \mathbf{Z}_{1, N/2} \\ \mathbf{Z}_{1, N/2} & \mathbf{P}_{N-1, N/2} \end{bmatrix},$$

where  $\mathbf{I}_{N/2}$  and  $\bar{\mathbf{I}}_{N/2}$  are, respectively, the identity and counter-identity matrices of order  $N/2$  and  $\mathbf{P}_{N-1, N/2}$  is an  $(N-1) \times (N/2)$  matrix whose row vectors are defined by:

$$\mathbf{P}_{N-1, N/2}^{(i)} = \begin{cases} \mathbf{Z}_{1, N/2}, & \text{if } i = 1, 3, 5, \dots, N-1 \\ \mathbf{I}_{N/2}^{(i/2)}, & \text{if } i = 0, 2, 4, \dots, N-2. \end{cases}$$

The scaling factor  $1/\sqrt{2}$  of (7) can be merged into the image/video compression quantization step. Furthermore, (1) can be applied to generate orthogonal versions of larger transforms. The computational cost of the resulting  $N$ -point transform is given by twice the number of bit-shifting operations of the original  $N/2$ -point transform; and twice the number of additions plus  $N$  extra additions. Following the described algorithm, we obtained the 16- and 32-point low complexity transform matrices proposed. More explicitly, we obtained the following 16- and 32-point matrices, respectively:

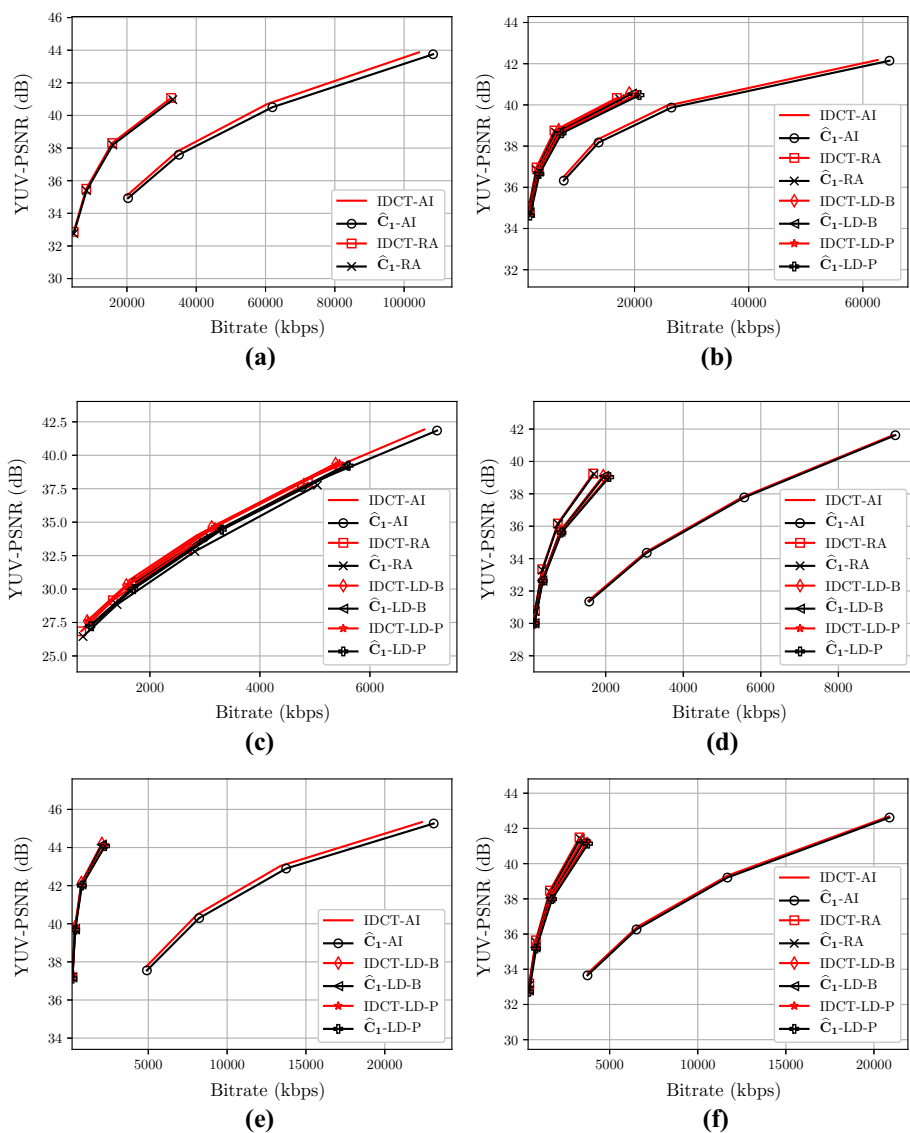
$$\mathbf{T}_{(16)} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 \\ 2 & 2 & 1 & 0 & 0 & -1 & -2 & -2 & -2 & -2 & -1 & 0 & 0 & 1 & 2 & 2 \\ 2 & 2 & 1 & 0 & 0 & -1 & -2 & -2 & -2 & -2 & 1 & 0 & 0 & -1 & -2 & -2 \\ 2 & 1 & -1 & -2 & -2 & -1 & 1 & 2 & 2 & 1 & -1 & -2 & -2 & -1 & 1 & 2 \\ 2 & 1 & -1 & -2 & -2 & -1 & 1 & 2 & -2 & -1 & 1 & 2 & 2 & 1 & -1 & -2 \\ 1 & 0 & -2 & -2 & 2 & 2 & 0 & -1 & -1 & 0 & 2 & 2 & -2 & -2 & 0 & 1 \\ 1 & 0 & -2 & -2 & 2 & 2 & 0 & -1 & 1 & 0 & -2 & -2 & 2 & 2 & 0 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 \\ 2 & -2 & 0 & 1 & -1 & 0 & 2 & -2 & 2 & 0 & -1 & 1 & 0 & -2 & 2 & 2 \\ 2 & -2 & 0 & 1 & -1 & 0 & 2 & -2 & 2 & 2 & 0 & 1 & -1 & 0 & 2 & -2 \\ 1 & -2 & 2 & -1 & -1 & 2 & -2 & 1 & 1 & -2 & 2 & -1 & -1 & 2 & -2 & 1 \\ 1 & -2 & 2 & -1 & -1 & 2 & -2 & 1 & 1 & -2 & 2 & -1 & -1 & 2 & -2 & -1 \\ 0 & -1 & 2 & -2 & 2 & -2 & 1 & 0 & 0 & 1 & -2 & 2 & -2 & 2 & -1 & 0 \\ 0 & -1 & 2 & -2 & 2 & -2 & 1 & 0 & 0 & 1 & -2 & 2 & -2 & 2 & -1 & 0 \end{bmatrix}$$

and

**Table 9** Computational cost comparison for 8-, 16-, and 32-point transforms embedded in HEVC reference software

| $N$ | IDCT (Meher et al. 2014) |            | Proposed transform |            |
|-----|--------------------------|------------|--------------------|------------|
|     | Additions                | Bit-shifts | Additions          | Bit-shifts |
| 8   | 50                       | 30         | 24                 | 6          |
| 16  | 186                      | 86         | 64                 | 12         |
| 32  | 682                      | 278        | 160                | 24         |





**Fig. 11** Rate distortion curves of the modified HEVC software for test sequences: **a** PeopleOnStreet, **b** BasketballDrive, **c** RaceHorses, **d** BlowingBubbles, **e** KristenAndSara, **f** BasketbalDrillText

**Table 10** BD-PSNR (dB) and BD-Rate (%) of the modified HEVC reference software for tested video sequences

| Video sequence      | AI      |          | RA      |          | LD-B    |          | LD-P    |          |
|---------------------|---------|----------|---------|----------|---------|----------|---------|----------|
|                     | BD-PSNR | BD-Rate  | BD-PSNR | BD-Rate  | BD-PSNR | BD-Rate  | BD-PSNR | BD-Rate  |
| PeopleOnStreet      | 0.2999  | − 5.5375 | 0.1467  | − 3.4323 | N/A     | N/A      | N/A     | N/A      |
| BasketballDrive     | 0.1692  | − 6.1033 | 0.1412  | − 6.1876 | 0.1272  | − 5.2730 | 0.1276  | − 5.2407 |
| RaceHorses          | 0.4714  | − 5.8250 | 0.5521  | − 8.6149 | 0.5460  | − 7.9067 | 0.5344  | − 7.6868 |
| BlowingBubbles      | 0.0839  | − 1.4715 | 0.0821  | − 2.1612 | 0.0806  | − 2.1619 | 0.0813  | − 2.2370 |
| KristenAndSara      | 0.2582  | − 5.0441 | N/A     | N/A      | 0.1230  | − 4.1823 | 0.1118  | − 4.0048 |
| BasketballDrillText | 0.1036  | − 1.9721 | 0.1372  | − 3.2741 | 0.1748  | − 4.3383 | 0.1646  | − 4.1509 |

 $\mathbf{T}_{(32)}$ 

[illegible]

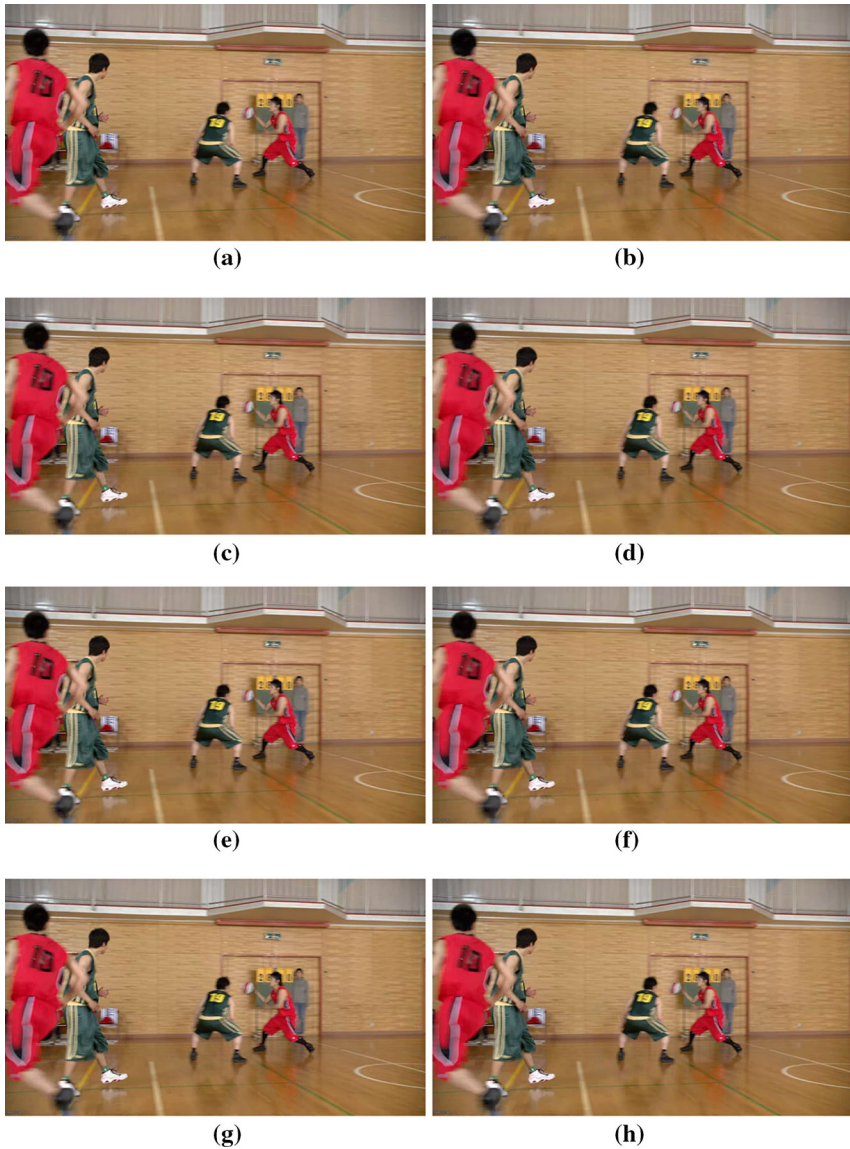
The resulting approximations for the above low-complexity matrices can be found from (1) and (2). The diagonal matrices implied by (2) are  $\mathbf{D}_{(16)} = 4 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \text{diag}(4, 9, 10, 9) \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  and  $\mathbf{D}_{(32)} = 2 \cdot \mathbf{D}_{(16)} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , respectively, where  $\otimes$  is the Kronecker product (Seber 2008).

Figures 9 and 10 display the SFG for the low-complexity transform matrices  $\mathbf{T}_{(16)}$  and  $\mathbf{T}_{(32)}$  derived from  $\mathbf{T}_1$ .

Table 9 lists the computational costs of the proposed transform for sizes  $N = 8, 16, 32$  compared to an efficient implementation of the IDCT (Meher et al. 2014).

In our experiments, the original 8-, 16-, and 32-point integer transforms of HEVC were substituted by  $\widehat{C}_1$  and its scaled versions. The original 4-point transform was kept unchanged because it is already a very low-complexity transformation. We encoded the first 100 frames of one video sequence of each A to F class in accordance with the common test conditions (CTC) documentation (Bossen 2013). Namely we used the 8-bit videos: PeopleOnStreet ( $2560 \times 1600$  at 30 fps), BasketballDrive ( $1920 \times 1080$  at 50 fps), RaceHorses ( $832 \times 480$  at 30 fps), BlowingBubbles ( $416 \times 240$  at 50 fps), KristenAndSara ( $1280 \times 720$  at 60 fps), and BasketballDrillText ( $832 \times 480$  at 50 fps). As suggested in Jridi et al. (2015), all the test parameters were set according to the CTC documentation. We tested the proposed transforms in All Intra (AI), Random Access (RA), Low Delay B (LD-B), and Low Delay P (LD-P) configurations, all in the Main profile.

We selected the frame-by-frame MSE and PSNR (Ohm et al. 2012) for each YUV color channel as figures of merit. Then, for all test videos, we computed the rate distortion (RD) curve considering the recommended quantization parameter (QP) values, i.e. 22, 27, 32,



**Fig. 12** Compression of the tenth frame of BasketballDrive using **a, c, e** the default and **b, d, f** the modified versions of the HEVC software for QP = 32, and AI, RA, LD-B, and LD-P coding configurations, respectively. **a** MSE-Y = 10.4097, MSE-U = 3.5872, MSE-V = 3.3079, PSNR-Y = 37.9564, PSNR-U = 42.5832, PSNR-V = 42.9353, **b** MSE-Y = 10.8159, MSE-U = 3.8290, MSE-V = 3.5766, PSNR-Y = 37.7902, PSNR-U = 42.2999, PSNR-V = 42.5961, **c** MSE-Y = 10.1479, MSE-U = 3.4765, MSE-V = 3.1724, PSNR-Y = 38.0670, PSNR-U = 42.7194, PSNR-V = 43.1170, **d** MSE-Y = 10.3570, MSE-U = 3.6228, MSE-V = 3.3113, PSNR-Y = 37.9785, PSNR-U = 42.5403, PSNR-V = 42.9308, **e** MSE-Y = 14.0693, MSE-U = 4.0741, MSE-V = 4.4404, PSNR-Y = 36.6481, PSNR-U = 42.0304, PSNR-V = 41.6566, **f** MSE-Y = 14.5953, MSE-U = 4.1377, MSE-V = 4.6053, PSNR-Y = 36.4887, PSNR-U = 41.9632, PSNR-V = 41.4982, **g** MSE-Y = 14.6155, MSE-U = 4.1349, MSE-V = 4.5502, PSNR-Y = 36.4827, PSNR-U = 41.9661, PSNR-V = 41.5505, **h** MSE-Y = 15.0761, MSE-U = 4.2812, MSE-V = 4.6444, PSNR-Y = 36.3479, PSNR-U = 41.8151, PSNR-V = 41.4615

and 37 (Bossen 2013). The resulting RD curves are depicted in Fig. 11. We have also measured the Bjøntegaard's delta PSNR (BD-PSNR) and delta rate (BD-Rate) (Bjøntegaard 2001; Hanhart and Ebrahimi 2014) for the modified HEVC software. These values are summarized in Table 10. We demonstrate that replacing the IDCT by the proposed transform and its scaled versions results in a loss in quality of at most 0.47 dB for the AI configuration, which corresponds to an increase of 5.82% in bitrate. Worst performance for the other configurations—RA, LD-B, and LD-P—are found for the *KristenAndSara* video sequence, where approximately 0.55 dB are lost if compared to the original HEVC implementation.

Despite the very low computational cost when compared to the IDCT (cf. Table 9), the proposed transform does not introduce significant errors. Figure 12 illustrates the tenth frame of the *BasketballDrive* video encoded according to the default HEVC IDCT and  $\hat{C}_1$  and its scaled versions for each coding configuration. The QP was set to 32. Visual degradations are virtually nonperceptible demonstrating real-world applicability of the proposed DCT approximations for high resolution video coding.

## 7 Conclusion

In this paper, we set up and solved an optimization problem aiming at the proposition of new approximations for the 8-point DCT. The obtained approximations were determined according to a greedy heuristic which minimized the angle between the rows of the approximate and the exact DCT matrices. Constraints of orthogonality and low computational complexity were imposed. *One of the obtained approximations outperformed all the considered approximations in literature according to popular performance measures.* We also introduced the use of circular statistics for assessing approximate transforms. For the proposed transform  $T_1$ , a fast algorithm requiring only 24 additions and 6 bit-shifting operations was proposed. The fast algorithm for the proposed method and directly competing approximations were given FPGA realizations. Simulations were made and the hardware resource consumption and power consumption were measured. The maximum operating frequency of the proposed method was 37.4% higher when compared with the well-known Lengwehasatit–Ortega approximation (LO) (Lengwehasatit and Ortega 2004). In addition, the applicability of the proposed approximation in the context of image compression and video coding was demonstrated. Our experiments also demonstrate that DCT approximations can effectively approximate the DCT behavior, but also—under particular conditions—outperform the DCT itself for image coding. The proposed approximation is fully HEVC-compliant, being capable of video coding with HEVC quality at lower computational costs.

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